

Problem 1)

$$\begin{cases} mv_i + MV_i = mv_f + MV_f \\ \frac{1}{2}mv_i^2 + \frac{1}{2}MV_i^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}MV_f^2 \end{cases}$$

$$\Rightarrow \begin{cases} m(v_i - v_f) = M(V_f - V_i) \\ m(v_i^2 - v_f^2) = M(V_f^2 - V_i^2) \end{cases} \quad \text{divide the two eqns.}$$

$$\Rightarrow v_i + v_f = V_i + V_f \Rightarrow v_i - V_i = -(v_f - V_f)$$

Problem 2)

a) $F = \frac{dP}{dt} = \frac{d(mv)}{dt} = m \frac{dv}{dt} + \frac{dm}{dt} v = 0 + \sigma v$

$$\Rightarrow F = \sigma v$$

b) $\frac{dT}{dt} = \frac{d}{dt} \left(\frac{1}{2}mv^2 \right) = \frac{dm}{dt} \left(\frac{v^2}{2} \right) = \frac{\sigma v^2}{2}$

c) $\frac{dW}{dt} = \frac{F dx}{dt} = F \cdot v = \sigma v^2$

d) Work-energy theorem states that, $\text{heat} = W - T$
 $= \sigma v^2 - \frac{\sigma v^2}{2} = \boxed{\frac{\sigma v^2}{2}}$

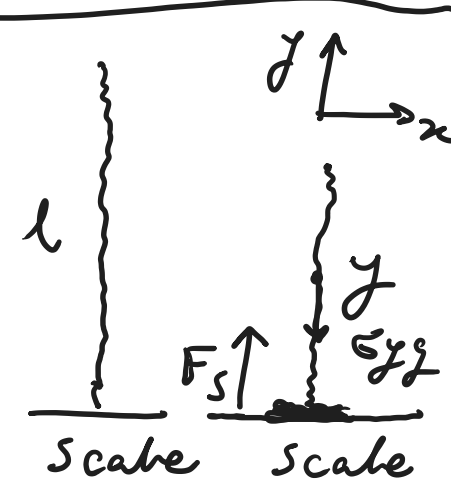
Problem 3)

$$\Sigma F = F_s - \sigma Lg = \frac{dP}{dt} = \frac{d(\sigma y \dot{y})}{dt} = \sigma y \ddot{y} + \sigma \dot{y}^2 \quad \text{①}$$

The part of chain in air is in free fall. $\Rightarrow \ddot{y} = -g \quad \text{②}$

From energy conservation we have, $\dot{y} = \sqrt{2g(L-y)} \quad \text{③}$

$$\text{① ② ③} \Rightarrow F = \sigma Lg - \sigma y g + 2\sigma(L-y)g = 3\sigma(L-y)g$$



Problem 4)

The horizontal velocity of m before and after collision is $-V$ and $V \cos 2\theta$. Thus, the net gain in momentum of m is $mV(1 + \cos 2\theta)$.

Now, we need to compute the at which this momentum is transferred to compute $\vec{F} = \frac{d\vec{P}}{dt}$ the drag force.

The ring area at angle θ is,

$$dA = (2\pi R \sin \theta)(R d\theta)$$

However, only the effective (orthogonal area to V) is relevant to computing the rate of collisions. Thus,

$$dA_{\text{eff}} = dA \cos \theta$$

Thus, the volume swept by M at speed V is,

$$dv = V dA_{\text{eff}} = V(2\pi R \sin \theta)(R d\theta) \cos \theta$$

The collision rate is,

$$dp = n dv$$

The differential drag force is,

$$dF(\theta) = \frac{dP}{dt} = mV(1 + \cos 2\theta) dp$$

$$\begin{aligned} \int_0^{\pi/2} dF(\theta) &= \bar{F} = \int_0^{\pi/2} n(2\pi R^2 \sin \theta \cos \theta) mV(1 + \cos 2\theta) d\theta \\ &= 2\pi n m R^2 V^2 \int_0^{\pi/2} \sin \theta \cos \theta (1 + \cos 2\theta) d\theta \\ &= \int_0^{\pi/2} \left(\frac{\sin 2\theta}{2} + \frac{\sin 4\theta}{4} \right) d\theta \\ &= \left(-\frac{\cos 2\theta}{4} - \frac{\cos 4\theta}{16} \right) \Big|_0^{\pi/2} \end{aligned}$$

$$\boxed{\bar{F} = \pi n m R^2 V^2}$$

